

Certified Analytical Bounds for Multi-Phase Hypersonic Reachable Sets

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Abstract

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1 Introduction

1.1 Grid-based Hamilton-Jacobi-Isaacs evaluations

1.2 Fixed-grid pseudospectral methods

1.3 Thesis statement

1.4 Outline of contributions

2 The Model Problem

2.1 Abstract formulation of the multi-phase optimal control problem

The vehicle is a rigid body, not a point mass (Appendix A). Its state

$$x = (r, \theta, \phi, u, v, w, q_0, q_1, q_2, q_3, \omega_1, \omega_2, \omega_3) \in \mathcal{X} \subseteq \mathbb{R}^{13}, \quad (2.1)$$

augmented by mass, recession depth s and modal state $(\eta, \dot{\eta})$ where the corresponding mode or coupling is active, carries geocentric position, body-frame velocity, attitude quaternion and body rates as components. The attitude coordinates are redundant: every admissible state satisfies

$$q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1, \quad (2.2)$$

so the thirteen coordinates carry twelve degrees of freedom. The control

$$c = (\vec{\delta}, \vec{\tau}_{\text{rcs}}, \varsigma) \in U, \quad (2.3)$$

collects flap deflections $\vec{\delta}$, reaction-control moments $\vec{\tau}_{\text{rcs}}$ and, in a propulsive mode, throttle ς ; $U \subset \mathbb{R}^{n_c}$ is compact.

Crucially the incidence angles (α, β) are *outputs* of the attitude solution (??), not commanded quantities: trim is not assumed solvable, and attitudes the vehicle cannot hold are excluded automatically rather than by fiat. The reachable set cannot contain a state reached only by an unattainable incidence.

list deflection, RCS, and throttle limits (one citation each). compactness follows.

elaborate

2.2 The mode alphabet

2.3 Trajectory families as admissible words

2.4 The entry corridor: hypersonic path constraints

2.5 Certified envelopes for constitutive and coupling effects

2.6 Phase boundaries: physical, constitutive, and numerical

2.7 Asymptotic regimes and the dimensionless groups

2.8 Quantity of interest and uncertainty

3 Infinite-Dimensional Embedding and Measure Theory

3.1 Sobolev space formulation

3.2 Occupation measures and the weak Liouville identity

3.3 Per-mode measures, guard measures, and linkage

3.4 The embedding as a linear problem on a Banach space

4 Concentration-Compactness and Profile Decomposition

- 4.1 Prokhorov tightness as a preliminary
- 4.2 The singular limit and the scaling group
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- 4.7 The compact residual
- 4.8 Consequence: dimension reduction for the certificate

5 Bounding Limits and Certificates

5.1 Relaxation and the direction of inclusion

5.2 The comparison principle

5.3 Constructing the certificate from the decomposition

5.4 Verification over the corridor

5.5 Exact arithmetic and machine-checkable certification

5.6 Inner point cloud and the bounding gap

5.7 Limits of the analytical construction

6 Verification and Results

- 6.1 Validating the degenerate limit
- 6.2 Tightness against the extremal cloud
- 6.3 Independent numerical cross-check
- 6.4 The corridor constraints as hypotheses
- 6.5 Multi-mode words
- 6.6 Numerical evidence for the profile decomposition
- 6.7 Computational cost
- 6.8 Reproducibility environment

7 Discussion

7.1 Limitations

7.2 Scope: the boost-glide and aeroassisted class

7.3 Extensions

References

A Six-Degree-of-Freedom Equations of Motion

This appendix records the rigid-body dynamics the body of the paper treats as given. Nothing here is a contribution; the derivation is standard entry mechanics assembled once with a fixed set of frames and a fixed parametrization. Frame errors are the classic 6-DOF defect and are invisible in the assembled algebra, so the transformations are stated explicitly and used without exception.

[cite sources](#)

A.1 State vector and frames

Frames. Four frames are used throughout.

- $\mathcal{I}$: Earth-centred inertial (ECI), non-rotating.
- $\mathcal{E}$: Earth-centred Earth-fixed (ECEF), rotating with the constant planetary rate $\vec{\omega}_E = \omega_E \hat{z}_{\mathcal{I}}$.
- $\mathcal{L}$: local geocentric frame at the vehicle, with unit vectors $(\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi)$ pointing radially outward, east, and north.
- $\mathcal{B}$: body frame, fixed at the instantaneous centre of mass, axes along the vehicle's principal design directions.

The attitude quaternion $\vec{q}$ carries $\mathcal{I} \rightarrow \mathcal{B}$; the geocentric angles (θ, ϕ) carry $\mathcal{I} \rightarrow \mathcal{L}$.

The direction-cosine matrix built from $\vec{q}$,

$$\mathbf{R}_{B/I}(\vec{q}) = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 + q_0q_1) \\ 2(q_1q_3 + q_0q_2) & 2(q_2q_3 - q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix}, \quad (\text{A.1})$$

is polynomial of degree two in $\vec{q}$, which is the property [Appendix C](#) exploits. The geocentric map is

$$\mathbf{R}_{L/I}(\theta, \phi) = \begin{pmatrix} \cos \phi \cos \theta & \cos \phi \sin \theta & \sin \phi \\ -\sin \theta & \cos \theta & 0 \\ -\sin \phi \cos \theta & -\sin \phi \sin \theta & \cos \phi \end{pmatrix}. \quad (\text{A.2})$$

A.2 Translational equations

A.3 Rotational equations

A.4 Aerodynamic forces and moments

A.5 Atmosphere

A.6 Per-mode forms

A.7 Boundary discipline

B Corridor Constraints and Certified Envelopes

B.1 The corridor constraints in polynomial form

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